Aryan Amarbhai Bijva

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Research Interests

Foundation model adaptation for dense regression in medical imaging, geometric deep learning for measurement tasks, graph neural networks for anatomical structure modeling, and robust AI for clinical deployment.

Education

Government Engineering College, Patan Sept 2023 – June 2027 (Expected)
Bachelor of Engineering in Computer Science & Engineering *CGPA: 8.2 / 10.0*

Relevant Coursework: Linear Algebra, Multivariable Calculus, Probability & Statistics, Design & Analysis of Algorithms, Deep Learning, Computer Vision, Medical Image Analysis.

Research Grants & Awards

CDAC GPU Grant Nov 2025 – Present
MeitY, Government of India *National Level*

Awarded high-performance compute resources (NVIDIA A100/H100 clusters) for medical foundation model training.

First student in institutional history to secure a national-level GPU grant for AI research.

Publications & Submissions

Aryan Bijva and Krunal Suthar, "Diagnosing Adaptation Failure Modes in Foundation Models for RNFL Thickness Prediction," in *Proceedings of the 13th Ophthalmic Medical Image Analysis (OMIA) Workshop, held in conjunction with MICCAI*, 2026. [**Oral Presentation**]

Research Experience

Undergraduate Student Researcher Dec 2025 – Present
Foundation Model Adaptation for Retinal Image Analysis *GEC Patan*

Investigated failure modes in adapting Vision Foundation Models (RETFound) for continuous 360-point retinal nerve fiber layer (RNFL) thickness regression from fundus and SLO images, isolating spatial quantization and variance collapse.

Formulated a cyclic Graph Neural Network (GNN) adapter over frozen backbone patch tokens, enforcing closed topological constraints (S^1) and continuous coordinate regression without dense mask decoders.

Designed a combined multi-quantile pinball and cyclic Sobolev smoothness loss, recovering diagnostic prediction variance (σ: $4.8\,\mu$m $\rightarrow$ $11.8\,\mu$m) across acoustic and optical dropout zones.

Validated cross-site robustness and geometric fairness across FairFedMed, Harvard AI Lab metadata, and the GRAPE dataset; presented findings as an oral and poster presentation at OMIA 13 (MICCAI 2026).

Undergraduate Researcher Jan 2025 – Nov 2025
Glaucoma Diagnostic Benchmarking & Geometric Analysis *GEC Patan*

Developed an automated glaucoma diagnostic pipeline benchmarking fundus of REFUGE and GOALS datasets, achieving 92% screening accuracy.

Engineered automated optic disc and cup segmentation scripts for reproducible diagnostic benchmarking and continuous vertical cup-to-disc ratio (vCDR) extraction.

Optimized feature projection backbones using parameter-efficient fine-tuning for low-latency clinical edge deployment.

Technical Skills

ML & Deep Learning: Foundation Models (ViT, RETFound), Graph Neural Networks (GNNs), Dense Regression, Geometric Deep Learning, Domain Adaptation, Quantile Regression.
Medical Imaging: Fundus Photography, OCT Analysis, DICOM/NIfTI Processing, Retinal Biomarker Extraction, Clinical Validation Metrics.
Scientific Computing: Python (PyTorch, PyTorch Geometric, NumPy, SciPy), MATLAB, LaTeX.
Infrastructure: Linux/HPC Cluster Management, NVIDIA A100/H100 Optimization, Git.